

PROJECT CHARTER: STUDYCIRCLE

1. Project Overview & Identification

- **Project Name:** StudyCircle
- **Project Focus:** A gamified, multiplayer, web-based study companion system designed to mitigate remote learning isolation and "cyberloafing."
- **Target Domain:** Educational Technology / Computer-Supported Collaborative Learning (CSCL).
- **Geographical Scope:** Metro Manila, Philippines.
- **Temporal Scope:** January 2026 – May 2027.

2. Business Case & Problem Statement

2.1 The Problem

Academic environments are leaning heavily into remote and hybrid configurations. While increasing accessibility, this transition has introduced deep psychological and behavioral bottlenecks:

- **Cyberloafing & Attention Span:** 80% to 90% of college students routinely procrastinate. Continuous smartphone access causes frequent concentration failures, and human attention drops drastically after 8 to 10 minutes of repetitive tasks (*Vigilance Decrement*).
- **Psychological Exhaustion:** Students studying without organized cognitive cycles face a 50% higher rate of mental tiredness than those using natural cognitive endurance frameworks.
- **The Isolation Gap:** Remote settings lack the "social currency" and externalized accountability found in real classrooms. This amplifies the *planning fallacy* (underestimating task duration) and decreases cognitive control.

2.2 The Solution: StudyCircle

StudyCircle closes the gap between study intention and actual behavior. By integrating behavioral economics (**Nudge Theory**), cognitive alignment (**Cognitive Load Theory**), and cultural norms (**Pakikisama**), the platform converts solitary, distracting digital sessions into highly disciplined, community-backed gamified sprints.

3. Project Objectives

3.1 General Objective

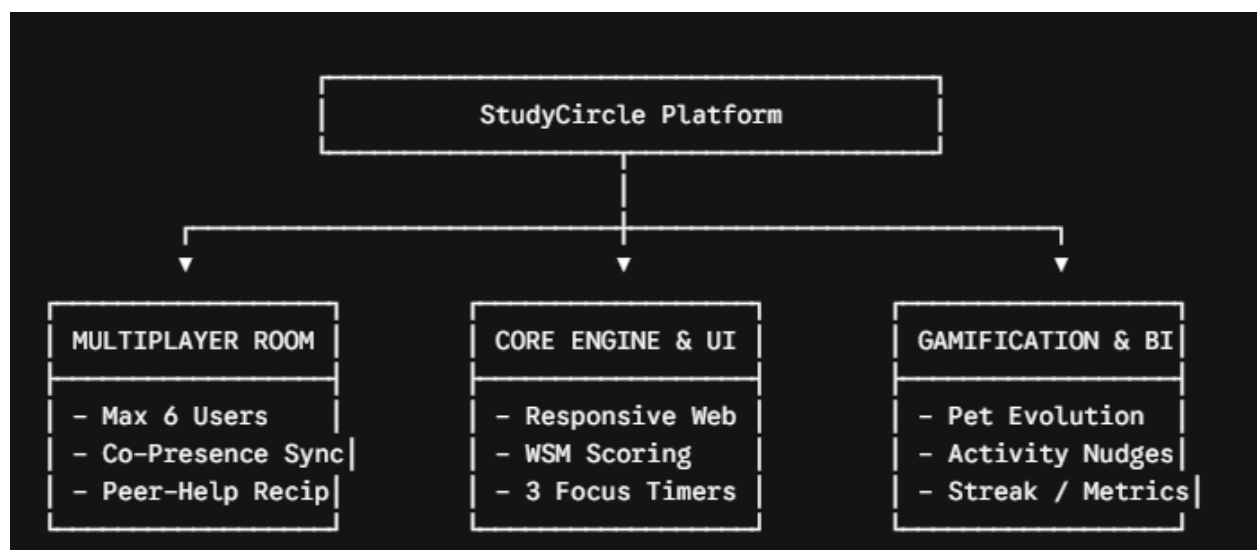
To develop **StudyCircle**, a gamified multiplayer web-based study companion system that utilizes structured study techniques and the **Weighted Sum Model (WSM)** algorithm to automate productivity scoring and foster social accountability, helping students maintain consistent and disciplined study habits in a remote environment.

3.2 Specific Project Milestones & Objectives

1. **Theoretical Alignment:** Research and embed the Pomodoro Technique, 52-17 Rule, and 90-Minute Ultradian Rhythm to address natural biological and focus rhythms.
2. **Gamification & Intervention Analysis:** Benchmark features using Nudge Theory, Flow Theory, and the *Tamagotchi Effect* via customizable virtual pets (Holland Lop, Golden Retriever, Siamese).
3. **Algorithmic Architecture:** Map out criteria parameters (duration, task completion, session consistency) to form data inputs for the analytical model.
4. **Scoring Configuration:** Implement the WSM formula to deliver objective, deterministic, and traceable productivity ratings and in-game reward payouts.
5. **Technical Production:** Construct a highly responsive, synchronized multiplayer platform equipped with real-time help systems, shared task trackers, anti-cyberloafing confirmation nudges, and detailed behavioral analytics dashboards.
6. **Quality Assurance (QA) Execution:** Validate functional mechanisms and test non-functional limits (WebSockets latency $\leq 200\text{ms}$; Database responses $\leq 100\text{ms}$; Uptime $\geq 99.9\%$).
7. **Standardized Assessment:** Evaluate the final platform layout and data integrity benchmarks using the ISO 25010 framework.
8. **Deployment:** Deliver and implement the operational production build.

4. Product Scope & Core Features

The system architecture must include the following functional components:



Multiplayer Virtual Rooms: Rooms capped at six users featuring avatar synchronization, real-time presence, and shared project lists to mimic physical classrooms.

- **Peer Help & Evaluation:** A live assistance request channel allowing students to help one another for bonus multipliers, backed by post-session peer productivity rankings.

- **Structured Timers:** Integrated system clocks dedicated exclusively to the Pomodoro, 52-17, and 90-minute ultradian cognitive cycles.
- **Interventionist Nudges:** Pop-up activity checks designed to safely disrupt cyberloafing habits without fracturing user workflow autonomy.
- **Gamified Rewards Systems:** Core token loop feeding into customizable digital avatars, housing assets, and responsive pet companions.
- **Data Analytics Engines:** Historical progress metrics compiling active streaks, cumulative hours, and weekly/monthly productivity trajectories.

5. Technology Stack & Technical Requirements

5.1 Software Engineering Environment

Layer	Technology	Operational Function
Backend Framework	Python-Flask	System architecture core, business logic, routing, and algorithmic handling.
Database Tier	SQLite + SQLAlchemy ORM	Local relational storage, data lineage tracking, and transactional user history.
Frontend Architecture	HTML5, JQuery, CSS, Bootstrap	Interface structure, client-side behaviors, and device-responsive adjustments.
Network & Protocols	WebSockets	Latency-compensated cross-room updates and state synchronization.

5.2 Performance & Security Guardrails (Non-Functional Requirements)

- **Synchronization Performance:** Keep timer synchronization delay strictly under **200ms** via WebSockets across room members.
- **Database Efficiency:** Keep underlying analytical queries and CRUD actions under **100ms**.
- **System Reliability:** Secure a total runtime operational uptime of **99.9%** with strict session persistence safeguards.

- **Cryptographic Standards:** Protect user credentials via **Bcrypt hashing** and restrict access tokens via **JSON Web Tokens (JWT)**.

6. Project Constraints & Assumptions

6.1 Constraints

- **Algorithmic Reliance:** System performance evaluation is completely tied to the logic and weight metrics assigned to the Weighted Sum Model (WSM). Any downstream bias or calibration errors will warp productivity values.
- **Geographic Boundaries:** Contextual validations, cultural parameter adjustments, and localized tests are customized solely for environments inside Metro Manila, Philippines.
- **Timeline Boundary:** Development ceases completely in May 2027; technological advancements emerging past this date are outside project parameters.

6.2 Assumptions

- Target users possess constant internet connection profiles stable enough to maintain real-time bidirectional WebSocket payloads.
- Users will engage accurately with post-session peer review windows to guarantee reliable, non-skewed subjective data entries.

7. Key Terms & Framework References

- **Weighted Sum Model (WSM):** The multi-criteria decision algorithm running calculations across distinct productivity variables to output objective points.
- **Pakikisama:** The cultural value of social togetherness engineered into multiplayer rooms to generate peer accountability.
- **Tamagotchi Effect:** The psychological bond established with digital pets used here to boost long-term habit building.
- **Cyberloafing:** The practice of using devices for non-academic distractions during dedicated focus windows.